DBMS

Demo data seeding

```
C:\Users\Rakshitha\Desktop\MediRecord\web>node scripts\seed-demo.js
◇ injected env (8) from .env.local

Demo data loaded:
  users           4
  doctors         2
  patients        8
  medicines       12
  appointments    11
  visits          8
  prescriptions   8
  bills           8
  bills_paid      5
  audit_log_rows  9

Logins:
  Admin         admin@medirecord.com          rkb@123
  Receptionist  reception@medirecord.com      password123
  Doctor        priya.sharma@medirecord.com   doctor@123   (General Medicine)
  Doctor        arvind.rao@medirecord.com     doctor@123   (Pulmonology)
```

1. All the tables and views

```
mysql> prompt pes2ug24cs918 [\d] >
PROMPT set to 'pes2ug24cs918 [\d] >'
pes2ug24cs918 [(none)] >use medirecord;
Database changed
pes2ug24cs918 [medirecord] >show tables;
+-----------------------+
| Tables_in_medirecord  |
+-----------------------+
| app_user              |
| appointment           |
| audit_log             |
| bill                  |
| doctor                |
| doctor_dashboard_view |
| doctor_revenue_summary |
| medicine              |
| patient               |
| prescription          |
| prescription_item     |
| role                  |
| visit                 |
+-----------------------+
13 rows in set (0.01 sec)
```

2. Table structures

```
pes2ug24cs918 [medirecord] >desc role;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| role_id   | int         | NO   | PRI | NULL    | auto_increment |
| role_name | varchar(30) | NO   | UNI | NULL    |                |
+-----------+-------------+------+-----+---------+----------------+
2 rows in set (0.00 sec)

pes2ug24cs918 [medirecord] >desc app_user;
+---------------+--------------+------+-----+-------------------+-------------------+
| Field         | Type         | Null | Key | Default           | Extra             |
+---------------+--------------+------+-----+-------------------+-------------------+
| user_id       | int          | NO   | PRI | NULL              | auto_increment    |
| name          | varchar(100) | NO   |     | NULL              |                   |
| email         | varchar(150) | NO   | UNI | NULL              |                   |
| password_hash | varchar(255) | NO   |     | NULL              |                   |
| role_id       | int          | NO   | MUL | NULL              |                   |
| created_at    | timestamp    | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+--------------+------+-----+-------------------+-------------------+
6 rows in set (0.00 sec)

pes2ug24cs918 [medirecord] >desc doctor;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| doctor_id       | int          | NO   | PRI | NULL    | auto_increment |
| user_id         | int          | NO   | UNI | NULL    |                |
| specialization  | varchar(100) | NO   |     | NULL    |                |
| license_no      | varchar(50)  | NO   | UNI | NULL    |                |
| years_experience | int         | NO   |     | 0       |                |
+-----------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

```
pes2ug24cs918 [medirecord] >desc patient;
+-------------+--------------+------+-----+-------------------+-------------------+
| Field       | Type         | Null | Key | Default           | Extra             |
+-------------+--------------+------+-----+-------------------+-------------------+
| patient_id  | int          | NO   | PRI | NULL              | auto_increment    |
| name        | varchar(100) | NO   |     | NULL              |                   |
| dob         | date         | NO   |     | NULL              |                   |
| gender      | varchar(10)  | NO   |     | NULL              |                   |
| phone       | varchar(15)  | NO   | MUL | NULL              |                   |
| address     | varchar(255) | YES  |     | NULL              |                   |
| blood_group | varchar(5)   | YES  |     | NULL              |                   |
| created_at  | timestamp    | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-------------+--------------+------+-----+-------------------+-------------------+
8 rows in set (0.00 sec)

pes2ug24cs918 [medirecord] >desc appointment;
+----------------+-------------+------+-----+-----------+----------------+
| Field          | Type        | Null | Key | Default   | Extra          |
+----------------+-------------+------+-----+-----------+----------------+
| appointment_id | int         | NO   | PRI | NULL      | auto_increment |
| patient_id     | int         | NO   | MUL | NULL      |                |
| doctor_id      | int         | NO   | MUL | NULL      |                |
| scheduled_at   | timestamp   | NO   | MUL | NULL      |                |
| status         | varchar(20) | NO   |     | scheduled |                |
| created_by     | int         | NO   | MUL | NULL      |                |
+----------------+-------------+------+-----+-----------+----------------+
6 rows in set (0.00 sec)
```

```
pes2ug24cs918 [medirecord] >desc visit;
+----------------+--------------+------+-----+-------------------+-------------------+
| Field          | Type         | Null | Key | Default           | Extra             |
+----------------+--------------+------+-----+-------------------+-------------------+
| visit_id       | int          | NO   | PRI | NULL              | auto_increment    |
| appointment_id | int          | NO   | UNI | NULL              |                   |
| vitals_bp      | varchar(15)  | YES  |     | NULL              |                   |
| vitals_temp    | decimal(4,1) | YES  |     | NULL              |                   |
| vitals_pulse   | int          | YES  |     | NULL              |                   |
| diagnosis_code | varchar(20)  | YES  | MUL | NULL              |                   |
| mongo_note_id  | varchar(50)  | YES  | MUL | NULL              |                   |
| created_at     | timestamp    | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+--------------+------+-----+-------------------+-------------------+
8 rows in set (0.00 sec)

pes2ug24cs918 [medirecord] >desc medicine;
+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| medicine_id | int           | NO   | PRI | NULL    | auto_increment |
| name        | varchar(150)  | NO   |     | NULL    |                |
| form        | varchar(20)   | NO   |     | NULL    |                |
| unit_price  | decimal(10,2) | NO   |     | NULL    |                |
| stock_qty   | int           | NO   |     | 0       |                |
+-------------+---------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

```
pes2ug24cs918 [medirecord] >desc prescription;
+-----------------+-----------+------+-----+-------------------+-------------------+
| Field           | Type      | Null | Key | Default           | Extra             |
+-----------------+-----------+------+-----+-------------------+-------------------+
| prescription_id | int       | NO   | PRI | NULL              | auto_increment    |
| visit_id        | int       | NO   | UNI | NULL              |                   |
| created_at      | timestamp | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----------------+-----------+------+-----+-------------------+-------------------+
3 rows in set (0.00 sec)

pes2ug24cs918 [medirecord] >desc prescription_item;
+---------------------+-------------+------+-----+---------+----------------+
| Field               | Type        | Null | Key | Default | Extra          |
+---------------------+-------------+------+-----+---------+----------------+
| prescription_item_id | int         | NO   | PRI | NULL    | auto_increment |
| prescription_id     | int         | NO   | MUL | NULL    |                |
| medicine_id         | int         | NO   | MUL | NULL    |                |
| dosage              | varchar(50) | NO   |     | NULL    |                |
| frequency           | varchar(50) | NO   |     | NULL    |                |
| duration_days       | int         | NO   |     | NULL    |                |
+---------------------+-------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

```
pes2ug24cs918 [medirecord] >desc bill;
+------------------+---------------+------+-----+-------------------+-------------------+
| Field            | Type          | Null | Key | Default           | Extra             |
+------------------+---------------+------+-----+-------------------+-------------------+
| bill_id          | int           | NO   | PRI | NULL              | auto_increment    |
| visit_id         | int           | NO   | UNI | NULL              |                   |
| consultation_fee | decimal(10,2) | NO   |     | 0.00              |                   |
| medicine_total   | decimal(10,2) | NO   |     | 0.00              |                   |
| tax              | decimal(10,2) | NO   |     | 0.00              |                   |
| total_amount     | decimal(10,2) | NO   |     | NULL              |                   |
| payment_status   | varchar(20)   | NO   |     | pending           |                   |
| generated_at     | timestamp     | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+---------------+------+-----+-------------------+-------------------+
8 rows in set (0.00 sec)

pes2ug24cs918 [medirecord] >desc audit_log;
+------------+-------------+------+-----+-------------------+-------------------+
| Field      | Type        | Null | Key | Default           | Extra             |
+------------+-------------+------+-----+-------------------+-------------------+
| audit_id   | int         | NO   | PRI | NULL              | auto_increment    |
| table_name | varchar(50) | NO   | MUL | NULL              |                   |
| record_id  | int         | NO   |     | NULL              |                   |
| action     | varchar(10) | NO   |     | NULL              |                   |
| changed_by | int         | YES  | MUL | NULL              |                   |
| changed_at | timestamp   | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| old_value  | json        | YES  |     | NULL              |                   |
| new_value  | json        | YES  |     | NULL              |                   |
+------------+-------------+------+-----+-------------------+-------------------+
8 rows in set (0.00 sec)
```

3. Constraints and foreign keys

```
pes2ug24cs918 [medirecord] >SHOW CREATE TABLE appointment\G
*************************** 1. row ***************************
       Table: appointment
Create Table: CREATE TABLE `appointment` (
  `appointment_id` int NOT NULL AUTO_INCREMENT,
  `patient_id` int NOT NULL,
  `doctor_id` int NOT NULL,
  `scheduled_at` timestamp NOT NULL,
  `status` varchar(20) NOT NULL DEFAULT 'scheduled',
  `created_by` int NOT NULL,
  PRIMARY KEY (`appointment_id`),
  KEY `created_by` (`created_by`),
  KEY `idx_appointment_scheduled_at` (`scheduled_at`),
  KEY `idx_appointment_doctor` (`doctor_id`),
  KEY `idx_appointment_patient` (`patient_id`),
  CONSTRAINT `appointment_ibfk_1` FOREIGN KEY (`patient_id`) REFERENCES `patient` (`patient_id`) ON DELETE RESTRICT,
  CONSTRAINT `appointment_ibfk_2` FOREIGN KEY (`doctor_id`) REFERENCES `doctor` (`doctor_id`) ON DELETE RESTRICT,
  CONSTRAINT `appointment_ibfk_3` FOREIGN KEY (`created_by`) REFERENCES `app_user` (`user_id`) ON DELETE RESTRICT,
  CONSTRAINT `appointment_chk_1` CHECK ((`status` in (_utf8mb4'scheduled',_utf8mb4'completed',_utf8mb4'cancelled')))
) ENGINE=InnoDB AUTO_INCREMENT=12 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
1 row in set (0.00 sec)
```

4. Indexes

```
pes2ug24cs918 [medirecord] >
pes2ug24cs918 [medirecord] >SHOW INDEX FROM appointment;
+-------------+------------+--------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table       | Non_unique | Key_name                       | Seq_in_index | Column_name  | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-------------+------------+--------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| appointment |          0 | PRIMARY                        |            1 | appointment_id | A         |          11 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| appointment |          1 | created_by                     |            1 | created_by   | A         |           1 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| appointment |          1 | idx_appointment_scheduled_at   |            1 | scheduled_at | A         |          11 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| appointment |          1 | idx_appointment_doctor         |            1 | doctor_id    | A         |           2 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| appointment |          1 | idx_appointment_patient        |            1 | patient_id   | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
+-------------+------------+--------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
5 rows in set (0.01 sec)
```

5. Alter

```
pes2ug24cs918 [medirecord] >ALTER TABLE patient ADD COLUMN emergency_contact VARCHAR(15);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

pes2ug24cs918 [medirecord] >desc patient;
+-------------------+--------------+------+-----+-------------------+-------------------+
| Field             | Type         | Null | Key | Default           | Extra             |
+-------------------+--------------+------+-----+-------------------+-------------------+
| patient_id        | int          | NO   | PRI | NULL              | auto_increment    |
| name              | varchar(100) | NO   |     | NULL              |                   |
| dob               | date         | NO   |     | NULL              |                   |
| gender            | varchar(10)  | NO   |     | NULL              |                   |
| phone             | varchar(15)  | NO   | MUL | NULL              |                   |
| address           | varchar(255) | YES  |     | NULL              |                   |
| blood_group       | varchar(5)   | YES  |     | NULL              |                   |
| created_at        | timestamp    | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| emergency_contact | varchar(15)  | YES  |     | NULL              |                   |
+-------------------+--------------+------+-----+-------------------+-------------------+
9 rows in set (0.00 sec)

pes2ug24cs918 [medirecord] >ALTER TABLE patient DROP COLUMN emergency_contact;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

6. Create , drop

```
pes2ug24cs918 [medirecord] >CREATE TABLE temp_demo (id INT PRIMARY KEY, note VARCHAR(30));
Query OK, 0 rows affected (0.05 sec)

pes2ug24cs918 [medirecord] >show tables;
+------------------------+
| Tables_in_medirecord   |
+------------------------+
| app_user               |
| appointment            |
| audit_log              |
| bill                   |
| doctor                 |
| doctor_dashboard_view  |
| doctor_revenue_summary |
| medicine               |
| patient                |
| prescription           |
| prescription_item      |
| role                   |
| temp_demo              |
| visit                  |
+------------------------+
14 rows in set (0.00 sec)
```

```
pes2ug24cs918 [medirecord] >DROP TABLE temp_demo;
Query OK, 0 rows affected (0.02 sec)

pes2ug24cs918 [medirecord] >show tables;
+------------------------+
| Tables_in_medirecord   |
+------------------------+
| app_user               |
| appointment            |
| audit_log              |
| bill                   |
| doctor                 |
| doctor_dashboard_view  |
| doctor_revenue_summary |
| medicine               |
| patient                |
| prescription           |
| prescription_item      |
| role                   |
| visit                  |
+------------------------+
13 rows in set (0.00 sec)
```

7. Insert statements

```
pes2ug24cs918 [medirecord] >START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

pes2ug24cs918 [medirecord] >INSERT INTO patient (name, dob, gender, phone, address, blood_group)
    -> VALUES ('Demo Patient', '1995-05-05', 'Female', '9000000001', 'Bengaluru', 'A+');
Query OK, 1 row affected (0.00 sec)

pes2ug24cs918 [medirecord] >INSERT INTO appointment (patient_id, doctor_id, scheduled_at, status, created_by)
    -> VALUES (LAST_INSERT_ID(), 1, NOW(), 'scheduled', 2);
Query OK, 1 row affected (0.00 sec)

pes2ug24cs918 [medirecord] >SELECT * FROM patient ORDER BY patient_id DESC LIMIT 1;
+------------+--------------+------------+--------+------------+-----------+-------------+---------------------+
| patient_id | name         | dob        | gender | phone      | address   | blood_group | created_at          |
+------------+--------------+------------+--------+------------+-----------+-------------+---------------------+
|          9 | Demo Patient | 1995-05-05 | Female | 9000000001 | Bengaluru | A+          | 2026-11-02 22:43:41 |
+------------+--------------+------------+--------+------------+-----------+-------------+---------------------+
1 row in set (0.00 sec)

pes2ug24cs918 [medirecord] >SELECT * FROM appointment ORDER BY appointment_id DESC LIMIT 1;
+----------------+------------+-----------+---------------------+-----------+------------+
| appointment_id | patient_id | doctor_id | scheduled_at        | status    | created_by |
+----------------+------------+-----------+---------------------+-----------+------------+
|             12 |          9 |         1 | 2026-11-02 22:43:41 | scheduled |          2 |
+----------------+------------+-----------+---------------------+-----------+------------+
1 row in set (0.00 sec)

pes2ug24cs918 [medirecord] >ROLLBACK;
Query OK, 0 rows affected (0.01 sec)
```

8. Users and roles

```
pes2ug24cs918 [medirecord] >SELECT * FROM role;
+---------+--------------+
| role_id | role_name    |
+---------+--------------+
|       1 | Admin        |
|       2 | Doctor       |
|       3 | Receptionist |
+---------+--------------+
3 rows in set (0.00 sec)

pes2ug24cs918 [medirecord] >SELECT user_id, name, email, role_id FROM app_user;
+---------+----------------+-----------------------------+---------+
| user_id | name           | email                       | role_id |
+---------+----------------+-----------------------------+---------+
|       1 | Admin          | admin@medirecord.com        |       1 |
|       2 | Reception Desk | reception@medirecord.com    |       3 |
|       3 | Priya Sharma   | priya.sharma@medirecord.com |       2 |
|       4 | Arvind Rao     | arvind.rao@medirecord.com   |       2 |
+---------+----------------+-----------------------------+---------+
4 rows in set (0.00 sec)
```

9. Doctors

```
pes2ug24cs918 [medirecord] >SELECT * FROM doctor;
+-----------+---------+------------------+------------+------------------+
| doctor_id | user_id | specialization   | license_no | years_experience |
+-----------+---------+------------------+------------+------------------+
|         1 |       3 | General Medicine | KA98765    |                8 |
|         2 |       4 | Pulmonology      | KA55120    |               14 |
+-----------+---------+------------------+------------+------------------+
2 rows in set (0.00 sec)
```

10. Patients

```
pes2ug24cs918 [medirecord] >SELECT patient_id, name, gender, dob, phone, blood_group FROM patient;
+------------+---------------+--------+------------+------------+-------------+
| patient_id | name          | gender | dob        | phone      | blood_group |
+------------+---------------+--------+------------+------------+-------------+
|          1 | Ravi Kumar    | Male   | 1990-04-12 | 9876543210 | O+          |
|          2 | Sneha Rao     | Female | 1985-09-23 | 9845012345 | B+          |
|          3 | Karthik Nair  | Male   | 1978-01-30 | 9741098765 | A+          |
|          4 | Anita Desai   | Female | 1996-06-18 | 9822011223 | AB+         |
|          5 | Mohammed Faiz | Male   | 1982-11-02 | 9900112233 | O-          |
|          6 | Lakshmi Iyer  | Female | 1965-03-27 | 9444055667 | B-          |
|          7 | Rohan Gupta   | Male   | 2001-08-09 | 9810099887 | A+          |
|          8 | Divya Menon   | Female | 1993-12-15 | 9605566778 | O+          |
+------------+---------------+--------+------------+------------+-------------+
8 rows in set (0.00 sec)
```

11. Medicines

```
pes2ug24cs918 [medirecord] >SELECT * FROM medicine;
+-------------+------------------------+--------+------------+-----------+
| medicine_id | name                   | form   | unit_price | stock_qty |
+-------------+------------------------+--------+------------+-----------+
|           1 | Paracetamol 500mg      | tablet |       2.00 |       500 |
|           2 | Amoxicillin 500mg      | tablet |      12.00 |       300 |
|           3 | Cetirizine 10mg        | tablet |       3.50 |       400 |
|           4 | Salbutamol Inhaler     | other  |      95.00 |       120 |
|           5 | Cough Syrup (Benadryl) | syrup  |      85.00 |       150 |
|           6 | Pantoprazole 40mg      | tablet |       8.00 |       250 |
|           7 | Ibuprofen 400mg        | tablet |       4.00 |       350 |
|           8 | Azithromycin 500mg     | tablet |      30.00 |       200 |
|           9 | Amlodipine 5mg         | tablet |       6.00 |       300 |
|          10 | Sumatriptan 50mg       | tablet |      25.00 |       100 |
|          11 | Montelukast 10mg       | tablet |      14.00 |       200 |
|          12 | Domperidone 10mg       | tablet |       3.00 |       250 |
+-------------+------------------------+--------+------------+-----------+
12 rows in set (0.02 sec)
```

12. Appointments

```
pes2ug24cs918 [medirecord] >SELECT * FROM appointment;
+----------------+------------+-----------+---------------------+-----------+------------+
| appointment_id | patient_id | doctor_id | scheduled_at        | status    | created_by |
+----------------+------------+-----------+---------------------+-----------+------------+
|              1 |          1 |         2 | 2026-10-24 22:32:01 | completed |          2 |
|              2 |          2 |         2 | 2026-10-25 22:32:01 | completed |          2 |
|              3 |          3 |         1 | 2026-10-26 22:32:01 | completed |          2 |
|              4 |          4 |         1 | 2026-10-27 22:32:01 | completed |          2 |
|              5 |          5 |         2 | 2026-10-28 22:32:01 | completed |          2 |
|              6 |          6 |         1 | 2026-10-29 22:32:01 | completed |          2 |
|              7 |          7 |         1 | 2026-10-30 22:32:01 | completed |          2 |
|              8 |          8 |         1 | 2026-10-31 22:32:01 | completed |          2 |
|              9 |          1 |         2 | 2026-11-03 18:32:01 | scheduled |          2 |
|             10 |          3 |         1 | 2026-11-03 22:32:01 | scheduled |          2 |
|             11 |          4 |         1 | 2026-11-04 18:32:01 | scheduled |          2 |
+----------------+------------+-----------+---------------------+-----------+------------+
11 rows in set (0.00 sec)
```

13. Visits

```
pes2ug24cs918 [medirecord] >SELECT visit_id, appointment_id, vitals_bp, vitals_temp, vitals_pulse, diagnosis_code FROM visit;
+----------+----------------+-----------+-------------+--------------+----------------+
| visit_id | appointment_id | vitals_bp | vitals_temp | vitals_pulse | diagnosis_code |
+----------+----------------+-----------+-------------+--------------+----------------+
|        1 |              1 | 128/82    |        98.4 |           88 | J45            |
|        2 |              2 | 118/76    |        98.6 |           82 | J45            |
|        3 |              3 | 122/80    |       101.2 |           96 | J06            |
|        4 |              4 | 110/70    |        98.2 |           78 | K29            |
|        5 |              5 | 136/88    |        98.8 |           92 | J44            |
|        6 |              6 | 160/100   |        98.4 |           84 | I10            |
|        7 |              7 | 120/78    |       100.4 |           90 | J06            |
|        8 |              8 | 112/72    |        98.0 |           74 | G43            |
+----------+----------------+-----------+-------------+--------------+----------------+
8 rows in set (0.02 sec)
```

14. Prescriptions and their items

```
pes2ug24cs918 [medirecord] >SELECT * FROM prescription;
+-----------------+----------+---------------------+
| prescription_id | visit_id | created_at          |
+-----------------+----------+---------------------+
|               1 |        1 | 2026-11-02 22:32:01 |
|               2 |        2 | 2026-11-02 22:32:01 |
|               3 |        3 | 2026-11-02 22:32:01 |
|               4 |        4 | 2026-11-02 22:32:01 |
|               5 |        5 | 2026-11-02 22:32:01 |
|               6 |        6 | 2026-11-02 22:32:01 |
|               7 |        7 | 2026-11-02 22:32:01 |
|               8 |        8 | 2026-11-02 22:32:01 |
+-----------------+----------+---------------------+
8 rows in set (0.00 sec)
```

```
pes2ug24cs918 [medirecord] >SELECT * FROM prescription_item;
+----------------------+-----------------+-------------+----------+----------------------+---------------+
| prescription_item_id | prescription_id | medicine_id | dosage   | frequency            | duration_days |
+----------------------+-----------------+-------------+----------+----------------------+---------------+
|                    1 |               1 |           4 | 2 puffs  | 4 times a day        |            14 |
|                    2 |               1 |          11 | 1 tablet | once at night        |            14 |
|                    3 |               2 |           4 | 2 puffs  | as needed            |            10 |
|                    4 |               2 |           3 | 1 tablet | once a day           |             7 |
|                    5 |               3 |           1 | 1 tablet | 3 times a day        |             3 |
|                    6 |               4 |           6 | 1 tablet | before breakfast     |            14 |
|                    7 |               4 |          12 | 1 tablet | twice a day          |             5 |
|                    8 |               5 |           4 | 2 puffs  | 3 times a day        |            30 |
|                    9 |               5 |           8 | 1 tablet | once a day           |             5 |
|                   10 |               6 |           9 | 1 tablet | once a day           |            30 |
|                   11 |               7 |           1 | 1 tablet | 3 times a day        |             3 |
|                   12 |               7 |           5 | 10 ml    | 3 times a day        |             5 |
|                   13 |               8 |          10 | 1 tablet | at onset of headache |             5 |
|                   14 |               8 |          12 | 1 tablet | as needed            |             5 |
+----------------------+-----------------+-------------+----------+----------------------+---------------+
14 rows in set (0.00 sec)
```

15. Bills

```
pes2ug24cs918 [medirecord] >SELECT * FROM bill;
+---------+----------+------------------+----------------+--------+--------------+----------------+---------------------+
| bill_id | visit_id | consultation_fee | medicine_total | tax    | total_amount | payment_status | generated_at        |
+---------+----------+------------------+----------------+--------+--------------+----------------+---------------------+
|       1 |        1 |           500.00 |        1526.00 | 101.30 |      2127.30 | paid           | 2026-11-02 22:32:01 |
|       2 |        2 |           500.00 |         974.50 |  73.73 |      1548.23 | paid           | 2026-11-02 22:32:01 |
|       3 |        3 |           300.00 |           6.00 |  15.30 |       321.30 | paid           | 2026-11-02 22:32:01 |
|       4 |        4 |           300.00 |         127.00 |  21.35 |       448.35 | pending        | 2026-11-02 22:32:01 |
|       5 |        5 |           500.00 |        3000.00 | 175.00 |      3675.00 | pending        | 2026-11-02 22:32:01 |
|       6 |        6 |           300.00 |         180.00 |  24.00 |       504.00 | paid           | 2026-11-02 22:32:01 |
|       7 |        7 |           300.00 |         431.00 |  36.55 |       767.55 | pending        | 2026-11-02 22:32:01 |
|       8 |        8 |           300.00 |         140.00 |  22.00 |       462.00 | paid           | 2026-11-02 22:32:01 |
+---------+----------+------------------+----------------+--------+--------------+----------------+---------------------+
8 rows in set (0.00 sec)
```

16. Row count

```
pes2ug24cs918 [medirecord] >SELECT 'patient' AS tbl, COUNT(*) AS n FROM patient
    -> UNION SELECT 'doctor', COUNT(*) FROM doctor
    -> UNION SELECT 'appointment', COUNT(*) FROM appointment
    -> UNION SELECT 'visit', COUNT(*) FROM visit
    -> UNION SELECT 'prescription_item', COUNT(*) FROM prescription_item
    -> UNION SELECT 'bill', COUNT(*) FROM bill;
+------------------+----+
| tbl              | n  |
+------------------+----+
| patient          |  8 |
| doctor           |  2 |
| appointment      | 11 |
| visit            |  8 |
| prescription_item | 14 |
| bill             |  8 |
+------------------+----+
6 rows in set (0.00 sec)
```

17. JOIN across four tables

```
pes2ug24cs918 [medirecord] >SELECT p.name AS patient, u.name AS doctor, v.diagnosis_code, b.total_amount
    -> FROM visit v
    -> JOIN appointment a ON a.appointment_id = v.appointment_id
    -> JOIN patient p ON p.patient_id = a.patient_id
    -> JOIN doctor d ON d.doctor_id = a.doctor_id
    -> JOIN app_user u ON u.user_id = d.user_id
    -> LEFT JOIN bill b ON b.visit_id = v.visit_id;
+--------------+-------------+----------------+--------------+
| patient      | doctor      | diagnosis_code | total_amount |
+--------------+-------------+----------------+--------------+
| Ravi Kumar   | Arvind Rao  | J45            |      2127.30 |
| Sneha Rao    | Arvind Rao  | J45            |      1548.23 |
| Karthik Nair | Priya Sharma | J06           |       321.30 |
| Anita Desai  | Priya Sharma | K29           |       448.35 |
| Mohammed Faiz | Arvind Rao | J44            |      3675.00 |
| Lakshmi Iyer | Priya Sharma | I10           |       504.00 |
| Rohan Gupta  | Priya Sharma | J06           |       767.55 |
| Divya Menon  | Priya Sharma | G43           |       462.00 |
+--------------+-------------+----------------+--------------+
8 rows in set (0.00 sec)
```

18. query with GROUP BY, aggregate and HAVING

```
pes2ug24cs918 [medirecord] >SELECT u.name AS doctor, COUNT(*) AS visits
    -> FROM visit v
    -> JOIN appointment a ON a.appointment_id = v.appointment_id
    -> JOIN doctor d ON d.doctor_id = a.doctor_id
    -> JOIN app_user u ON u.user_id = d.user_id
    -> GROUP BY d.doctor_id
    -> HAVING COUNT(*) >= 4;
+--------------+--------+
| doctor       | visits |
+--------------+--------+
| Priya Sharma |      5 |
+--------------+--------+
1 row in set (0.00 sec)
```

19. ORDER BY with an aggregate

```
pes2ug24cs918 [medirecord] >SELECT payment_status, COUNT(*) AS bills, SUM(total_amount) AS amount
    -> FROM bill
    -> GROUP BY payment_status
    -> ORDER BY amount DESC;
+----------------+-------+---------+
| payment_status | bills | amount  |
+----------------+-------+---------+
| paid           |     5 | 4962.83 |
| pending        |     3 | 4890.90 |
+----------------+-------+---------+
2 rows in set (0.00 sec)
```

20. UPDATE

```
pes2ug24cs918 [medirecord] >START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

pes2ug24cs918 [medirecord] >UPDATE bill SET payment_status = 'paid' WHERE bill_id = 4;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

pes2ug24cs918 [medirecord] >SELECT bill_id, payment_status FROM bill WHERE bill_id = 4;
+---------+----------------+
| bill_id | payment_status |
+---------+----------------+
|       4 | paid           |
+---------+----------------+
1 row in set (0.00 sec)

pes2ug24cs918 [medirecord] >ROLLBACK;
Query OK, 0 rows affected (0.00 sec)
```

21. DELETE

```
pes2ug24cs918 [medirecord] >START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

pes2ug24cs918 [medirecord] >DELETE FROM medicine WHERE medicine_id NOT IN (SELECT medicine_id FROM prescription_item);
Query OK, 2 rows affected (0.00 sec)

pes2ug24cs918 [medirecord] >SELECT COUNT(*) AS medicines_left FROM medicine;
+----------------+
| medicines_left |
+----------------+
|             10 |
+----------------+
1 row in set (0.00 sec)

pes2ug24cs918 [medirecord] >ROLLBACK;
Query OK, 0 rows affected (0.01 sec)

pes2ug24cs918 [medirecord] >SELECT COUNT(*) AS medicines_after_rollback FROM medicine;
+--------------------------+
| medicines_after_rollback |
+--------------------------+
|                       12 |
+--------------------------+
1 row in set (0.00 sec)
```

22. correlated query

```
pes2ug24cs918 [medirecord] >SELECT b.bill_id, b.total_amount
    -> FROM bill b
    -> WHERE b.total_amount > (SELECT AVG(b2.total_amount) FROM bill b2 WHERE b2.visit_id <> b.visit_id);
+---------+--------------+
| bill_id | total_amount |
+---------+--------------+
|       1 |      2127.30 |
|       2 |      1548.23 |
|       5 |      3675.00 |
+---------+--------------+
3 rows in set (0.00 sec)
```

23. Nested query

```
pes2ug24cs918 [medirecord] >SELECT name FROM patient
    -> WHERE patient_id IN (SELECT patient_id FROM appointment WHERE status = 'scheduled');
+---------------+
| name          |
+---------------+
| Ravi Kumar    |
| Karthik Nair  |
| Anita Desai   |
+---------------+
3 rows in set (0.00 sec)
```

24. LEFT JOIN

```
pes2ug24cs918 [medirecord] >SELECT p.name FROM patient p
    -> LEFT JOIN appointment a ON a.patient_id = p.patient_id
    -> WHERE a.appointment_id IS NULL;
Empty set (0.00 sec)

pes2ug24cs918 [medirecord] >
```

Patient with no appointment is 0.

25. VIEWS

```
pes2ug24cs918 [medirecord] >SELECT * FROM doctor_revenue_summary;
+------------+---------------+--------------------+------------------+---------------+
| doctor_id | doctor_name   | total_appointments | unique_patients | total_revenue |
+------------+---------------+--------------------+------------------+---------------+
|          1 | Priya Sharma  |                  7 |                5 |       1287.30 |
|          2 | Arvind Rao    |                  4 |                3 |       3675.53 |
+------------+---------------+--------------------+------------------+---------------+
2 rows in set (0.01 sec)
```

```
pes2ug24cs918 [medirecord] >SELECT * FROM doctor_dashboard_view;
+-----------+--------------+----------------+----------------+------------+--------------+---------------------+--------------------+----------+----------------+----------------------------------+--------------+----------------+
| doctor_id | doctor_name  | specialization | appointment_id | patient_id | patient_name | scheduled_at        | appointment_status | visit_id | diagnosis_code | mongo_note_id                    | total_amount | payment_status |
+-----------+--------------+----------------+----------------+------------+--------------+---------------------+--------------------+----------+----------------+----------------------------------+--------------+----------------+
|         1 | Priya Sharma | General Medicine |              3 |          3 | Karthik Nair | 2026-10-26 22:32:01 | completed          |        3 | J06            | 6ae8c2093b65975e22e75a32         |       321.30 | paid           |
|         1 | Priya Sharma | General Medicine |              4 |          4 | Anita Desai  | 2026-10-27 22:32:01 | completed          |        4 | K29            | 6ae8c2093b65975e22e75a33         |       448.35 | pending        |
|         1 | Priya Sharma | General Medicine |              6 |          6 | Lakshmi Iyer | 2026-10-29 22:32:01 | completed          |        6 | I10            | 6ae8c2093b65975e22e75a35         |       504.00 | paid           |
|         1 | Priya Sharma | General Medicine |              7 |          7 | Rohan Gupta  | 2026-10-30 22:32:01 | completed          |        7 | J06            | 6ae8c2093b65975e22e75a36         |       767.55 | pending        |
|         1 | Priya Sharma | General Medicine |              8 |          8 | Divya Menon  | 2026-10-31 22:32:01 | completed          |        8 | G43            | 6ae8c2093b65975e22e75a37         |       462.00 | paid           |
|         1 | Priya Sharma | General Medicine |             10 |          3 | Karthik Nair | 2026-11-03 22:32:01 | scheduled          |     NULL | NULL           | NULL                             |         NULL | NULL           |
|         1 | Priya Sharma | General Medicine |             11 |          4 | Anita Desai  | 2026-11-04 18:32:01 | scheduled          |     NULL | NULL           | NULL                             |         NULL | NULL           |
|         2 | Arvind Rao   | Pulmonology    |              1 |          1 | Ravi Kumar   | 2026-10-24 22:32:01 | completed          |        1 | J45            | 6ae8c2093b65975e22e75a30         |      2127.30 | paid           |
|         2 | Arvind Rao   | Pulmonology    |              2 |          2 | Sneha Rao    | 2026-10-25 22:32:01 | completed          |        2 | J45            | 6ae8c2093b65975e22e75a31         |      1548.23 | paid           |
|         2 | Arvind Rao   | Pulmonology    |              5 |          5 | Mohammed Faiz | 2026-10-28 22:32:01 | completed          |        5 | J44            | 6ae8c2093b65975e22e75a34         |      3675.00 | pending        |
|         2 | Arvind Rao   | Pulmonology    |              9 |          1 | Ravi Kumar   | 2026-11-03 18:32:01 | scheduled          |     NULL | NULL           | NULL                             |         NULL | NULL           |
+-----------+--------------+----------------+----------------+------------+--------------+---------------------+--------------------+----------+----------------+----------------------------------+--------------+----------------+
11 rows in set (0.00 sec)
```

26. AUDIT LOG

```
pes2ug24cs918 [medirecord] >SELECT audit_id, table_name, action, changed_at FROM audit_log;
+----------+--------------+--------+---------------------+
| audit_id | table_name   | action | changed_at          |
+----------+--------------+--------+---------------------+
|        1 | prescription | INSERT | 2026-11-02 22:32:01 |
|        2 | prescription | INSERT | 2026-11-02 22:32:01 |
|        3 | prescription | INSERT | 2026-11-02 22:32:01 |
|        4 | prescription | INSERT | 2026-11-02 22:32:01 |
|        5 | visit        | UPDATE | 2026-11-02 22:32:01 |
|        6 | prescription | INSERT | 2026-11-02 22:32:01 |
|        7 | prescription | INSERT | 2026-11-02 22:32:01 |
|        8 | prescription | INSERT | 2026-11-02 22:32:01 |
|        9 | prescription | INSERT | 2026-11-02 22:32:01 |
+----------+--------------+--------+---------------------+
9 rows in set (0.02 sec)
```

27. TRIGGERS in db

```
pes2ug24cs918 [medirecord] >SELECT TRIGGER_NAME, EVENT_MANIPULATION AS event, EVENT_OBJECT_TABLE AS tbl, ACTION_TIMING AS timing
    -> FROM information_schema.TRIGGERS WHERE TRIGGER_SCHEMA = 'medirecord';
+------------------------+--------+-------------+--------+
| TRIGGER_NAME           | event  | tbl         | timing |
+------------------------+--------+-------------+--------+
| trg_prescription_insert | INSERT | prescription | AFTER  |
| trg_prescription_update | UPDATE | prescription | AFTER  |
| trg_prescription_delete | DELETE | prescription | AFTER  |
| trg_visit_update       | UPDATE | visit       | AFTER  |
+------------------------+--------+-------------+--------+
4 rows in set (0.03 sec)
```

28. TRIGGER

```
pes2ug24cs918 [medirecord] >SHOW CREATE TRIGGER trg_visit_update\G
*************************** 1. row ***************************
             Trigger: trg_visit_update
            sql_mode: ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION
SQL Original Statement: CREATE DEFINER=`root`@`localhost` TRIGGER `trg_visit_update` AFTER UPDATE ON `visit` FOR EACH ROW BEGIN
    IF NOT (OLD.diagnosis_code <=> NEW.diagnosis_code) THEN
        INSERT INTO audit_log (table_name, record_id, action, old_value, new_value)
        VALUES ('visit', NEW.visit_id, 'UPDATE',
                JSON_OBJECT('diagnosis_code', OLD.diagnosis_code),
                JSON_OBJECT('diagnosis_code', NEW.diagnosis_code));
    END IF;
END
  character_set_client: utf8mb4
  collation_connection: utf8mb4_0900_ai_ci
    Database Collation: utf8mb4_0900_ai_ci
               Created: 2026-10-22 23:54:45.12
1 row in set (0.00 sec)
```

29. Trigger firing live

```
pes2ug24cs918 [medirecord] >START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

pes2ug24cs918 [medirecord] >UPDATE visit SET diagnosis_code = 'J45' WHERE visit_id = 5;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

pes2ug24cs918 [medirecord] >SELECT audit_id, table_name, action FROM audit_log ORDER BY audit_id DESC LIMIT 2;
+----------+--------------+--------+
| audit_id | table_name   | action |
+----------+--------------+--------+
|       10 | visit        | UPDATE |
|        9 | prescription | INSERT |
+----------+--------------+--------+
2 rows in set (0.00 sec)

pes2ug24cs918 [medirecord] >ROLLBACK;
Query OK, 0 rows affected (0.01 sec)
```

30. Stored pocedure code

```
pes2ug24cs918 [medirecord] >SHOW CREATE PROCEDURE generate_bill\G
*************************** 1. row ***************************
           Procedure: generate_bill
            sql_mode: ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION
    Create Procedure: CREATE DEFINER=`root`@`localhost` PROCEDURE `generate_bill`(
    IN p_visit_id INT,
    IN p_consultation_fee DECIMAL(10,2),
    IN p_tax_rate DECIMAL(5,4)
)
BEGIN
    DECLARE v_medicine_total DECIMAL(10,2) DEFAULT 0;
    DECLARE v_tax DECIMAL(10,2);
    DECLARE v_total DECIMAL(10,2);

    SELECT COALESCE(SUM(m.unit_price * pi.duration_days), 0)
    INTO v_medicine_total
    FROM prescription p
    JOIN prescription_item pi ON pi.prescription_id = p.prescription_id
    JOIN medicine m ON m.medicine_id = pi.medicine_id
    WHERE p.visit_id = p_visit_id;

    SET v_tax = (p_consultation_fee + v_medicine_total) * p_tax_rate;
    SET v_total = p_consultation_fee + v_medicine_total + v_tax;

    INSERT INTO bill (visit_id, consultation_fee, medicine_total, tax, total_amount, payment_status)
    VALUES (p_visit_id, p_consultation_fee, v_medicine_total, v_tax, v_total, 'pending');

    SELECT LAST_INSERT_ID() AS new_bill_id;
END
character_set_client: utf8mb4
collation_connection: utf8mb4_0900_ai_ci
  Database Collation: utf8mb4_0900_ai_ci
1 row in set (0.00 sec)
```

31. Caling procedure inside a transaction

```
pes2ug24cs918 [medirecord] >START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

pes2ug24cs918 [medirecord] >INSERT INTO appointment (patient_id, doctor_id, scheduled_at, status, created_by)
    -> VALUES (1, 1, NOW(), 'completed', 2);
Query OK, 1 row affected (0.00 sec)

pes2ug24cs918 [medirecord] >INSERT INTO visit (appointment_id, vitals_bp, vitals_temp, vitals_pulse, diagnosis_code)
    -> VALUES (LAST_INSERT_ID(), '120/80', 98.6, 72, 'J06');
Query OK, 1 row affected (0.00 sec)

pes2ug24cs918 [medirecord] >CALL generate_bill(LAST_INSERT_ID(), 500, 0.05);
+-------------+
| new_bill_id |
+-------------+
|           9 |
+-------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

pes2ug24cs918 [medirecord] >SELECT bill_id, consultation_fee, medicine_total, tax, total_amount FROM bill ORDER BY bill_id DESC LIMIT 1;
+---------+------------------+----------------+-------+--------------+
| bill_id | consultation_fee | medicine_total | tax   | total_amount |
+---------+------------------+----------------+-------+--------------+
|       9 |           500.00 |           0.00 | 25.00 |       525.00 |
+---------+------------------+----------------+-------+--------------+
1 row in set (0.00 sec)

pes2ug24cs918 [medirecord] >ROLLBACK;
Query OK, 0 rows affected (0.00 sec)
```

```
pes2ug24cs918 [medirecord] >SELECT COUNT(*) AS bills_after_rollback FROM bill;
+----------------------+
| bills_after_rollback |
+----------------------+
|                    8 |
+----------------------+
1 row in set (0.00 sec)
```

32. Index used by query

```
pes2ug24cs918 [medirecord] >EXPLAIN SELECT appointment_id FROM appointment WHERE doctor_id = 1\G
*************************** 1. row ***************************
           id: 1
  select_type: SIMPLE
        table: appointment
   partitions: NULL
         type: ref
possible_keys: idx_appointment_doctor
          key: idx_appointment_doctor
      key_len: 4
          ref: const
         rows: 7
     filtered: 100.00
        Extra: Using index
1 row in set, 1 warning (0.00 sec)
```

33. Integrity errors

```
pes2ug24cs918 [medirecord] >INSERT INTO patient (name, dob, gender, phone) VALUES ('X', '2000-01-01', 'Alien', '1');
ERROR 3819 (HY000): Check constraint 'patient_chk_1' is violated.
pes2ug24cs918 [medirecord] >INSERT INTO medicine (name, form, unit_price, stock_qty) VALUES ('Bad Price', 'tablet', -5, 10);
ERROR 3819 (HY000): Check constraint 'medicine_chk_2' is violated.
pes2ug24cs918 [medirecord] >INSERT INTO app_user (name, email, password_hash, role_id) VALUES ('Dup', 'admin@medirecord.com', 'x', 1);
ERROR 1062 (23000): Duplicate entry 'admin@medirecord.com' for key 'app_user.email'
pes2ug24cs918 [medirecord] >INSERT INTO bill (visit_id, consultation_fee, medicine_total, tax, total_amount) VALUES (1, 100, 0, 0, 100);
ERROR 1062 (23000): Duplicate entry '1' for key 'bill.visit_id'
pes2ug24cs918 [medirecord] >DELETE FROM patient WHERE patient_id = 1;
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`medirecord`.`appointment`, CONSTRAINT `appointment_ibfk_1
` FOREIGN KEY (`patient_id`) REFERENCES `patient` (`patient_id`) ON DELETE RESTRICT)
pes2ug24cs918 [medirecord] >
```

```
C:\Users\Rakshitha\Desktop\MediRecord>mysql -u root -p --force medirecord < sql\demo_integrity_errors.sql
Enter password: *******
expected_to_fail
1. CHECK: invalid gender
ERROR 3819 (HY000) at line 16: Check constraint 'patient_chk_1' is violated.
expected_to_fail
2. CHECK: negative medicine price
ERROR 3819 (HY000) at line 20: Check constraint 'medicine_chk_2' is violated.
expected_to_fail
3. UNIQUE: duplicate login email
ERROR 1062 (23000) at line 24: Duplicate entry 'admin@medirecord.com' for key 'app_user.email'
expected_to_fail
4. UNIQUE: second bill for the same visit
ERROR 1062 (23000) at line 29: Duplicate entry '1' for key 'bill.visit_id'
expected_to_fail
5. FOREIGN KEY: appointment for non-existent patient
ERROR 1452 (23000) at line 33: Cannot add or update a child row: a foreign key constraint fails (`medirecord`.`appointme
nt`, CONSTRAINT `appointment_ibfk_1` FOREIGN KEY (`patient_id`) REFERENCES `patient` (`patient_id`) ON DELETE RESTRICT)
expected_to_fail
6. RESTRICT: delete a patient with appointments
ERROR 1451 (23000) at line 38: Cannot delete or update a parent row: a foreign key constraint fails (`medirecord`.`appoi
ntment`, CONSTRAINT `appointment_ibfk_1` FOREIGN KEY (`patient_id`) REFERENCES `patient` (`patient_id`) ON DELETE RESTRI
CT)
expected_to_fail
7. CHECK: invalid appointment status
ERROR 3819 (HY000) at line 42: Check constraint 'appointment_chk_1' is violated.
result
Done. Every statement above was rejected; data unchanged.
```

34. Password hashing

```
pes2ug24cs918 [medirecord] >SELECT email, LENGTH(password_hash) AS hash_length, LEFT(password_hash, 7) AS prefix FROM app_user;
+-----------------------------+-------------+---------+
| email                       | hash_length | prefix  |
+-----------------------------+-------------+---------+
| admin@medirecord.com        |          60 | $2b$10$ |
| reception@medirecord.com    |          60 | $2b$10$ |
| priya.sharma@medirecord.com |          60 | $2b$10$ |
| arvind.rao@medirecord.com   |          60 | $2b$10$ |
+-----------------------------+-------------+---------+
4 rows in set (0.00 sec)
```

35. Collections nd indexes

```
medirecord> db.clinical_notes.countDocuments()
8
medirecord> db.clinical_notes.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'idx_notes_text_search',
    weights: { notes: 1, symptoms: 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  },
  { v: 2, key: { visit_id: 1 }, name: 'visit_id_1', unique: true },
  { v: 2, key: { doctor_id: 1 }, name: 'doctor_id_1' },
  { v: 2, key: { diagnosis_code: 1 }, name: 'diagnosis_code_1' }
]
medirecord>
```

36. Data model

```
medirecord> db.clinical_notes.findOne({}, {_id:0})
{
  visit_id: 1,
  patient_id: 1,
  doctor_id: 2,
  diagnosis_code: 'J45',
  symptoms: [ 'chest tightness', 'wheezing', 'shortness of breath' ],
  notes: 'Patient reports wheezing at night and chest tightness on exertion. Known asthma since childhood. Reduced peak
flow. Advised to avoid dust exposure.',
  created_at: ISODate('2026-11-02T17:02:01.618Z')
}
medirecord>
```

37. Ranked full text search

```
medirecord> db.clinical_notes.find({ $text: { $search: "chest tightness wheezing" } },
{ _id:0, visit_id:1, symptoms:1, diagnosis_code:1, score:{ $meta:"textScore" } }).sort(
{ score:{ $meta:"textScore" } })
[
  {
    visit_id: 1,
    diagnosis_code: 'J45',
    symptoms: [ 'chest tightness', 'wheezing', 'shortness of breath' ],
    score: 4.083333333333333
  },
  {
    visit_id: 2,
    diagnosis_code: 'J45',
    symptoms: [ 'wheezing', 'dry cough', 'chest tightness' ],
    score: 3.0357142857142856
  },
  {
    visit_id: 5,
    diagnosis_code: 'J44',
    symptoms: [ 'shortness of breath', 'chronic cough', 'wheezing' ],
    score: 1.5333333333333332
  }
]
medirecord>
```

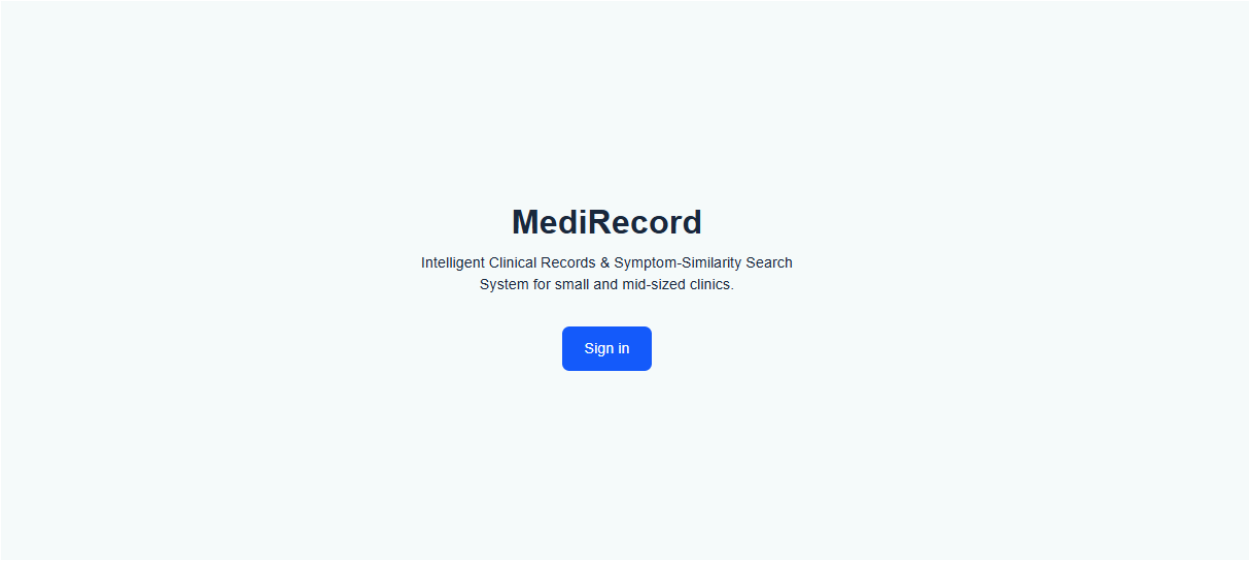
38. Filter bydiagnosis

```
medirecord> db.clinical_notes.find({ diagnosis_code: "J45" }, { _id:0, visit_id:1, doct
doctor_id:1, symptoms:1 })
[
  {
    visit_id: 1,
    doctor_id: 2,
    symptoms: [ 'chest tightness', 'wheezing', 'shortness of breath' ]
  },
  {
    visit_id: 2,
    doctor_id: 2,
    symptoms: [ 'wheezing', 'dry cough', 'chest tightness' ]
  }
]
```

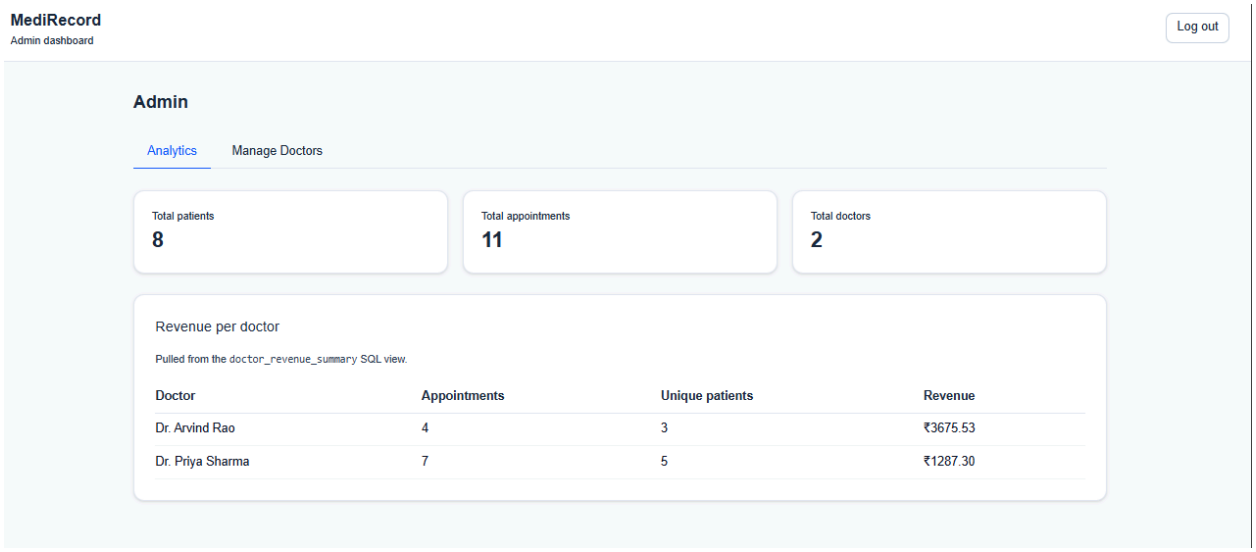
39. Aggregation notes per doc and diagnosis

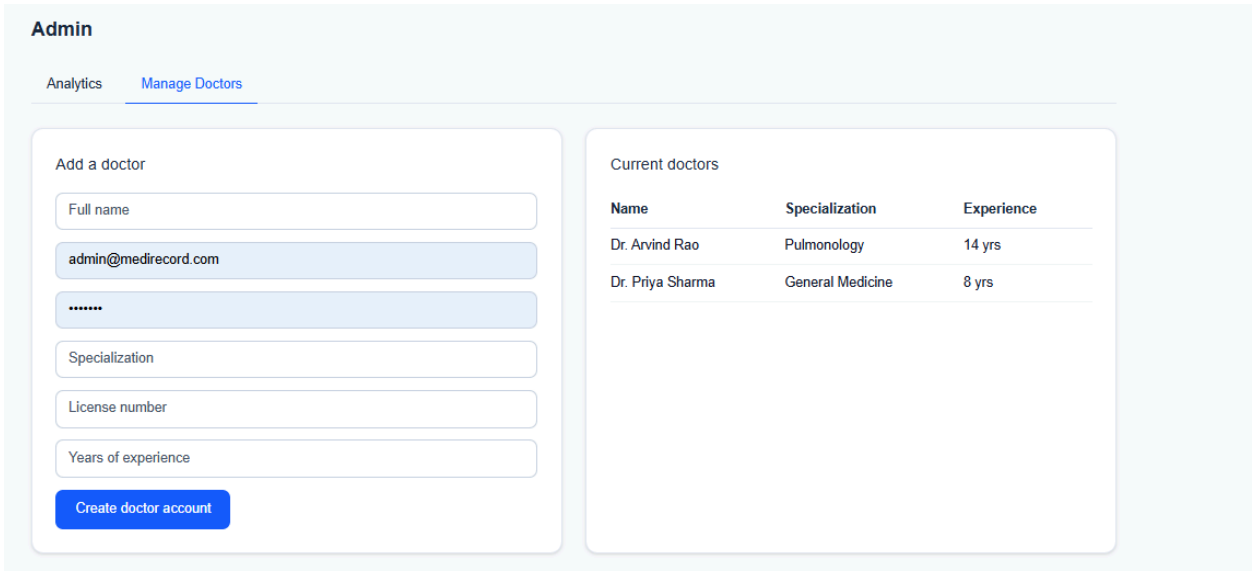
```
medirecord> db.clinical_notes.aggregate([ { $group: { _id: "$doctor_id", notes: { $sum: 1 } } },
{ $sort: { _id: 1 } } ])
[ { _id: 1, notes: 5 }, { _id: 2, notes: 3 } ]
medirecord> db.clinical_notes.aggregate([ { $group: { _id: "$doctor_id", notes: { $sum: 1 } } },
{ $sort: { _id: 1 } } ])
[ { _id: 1, notes: 5 }, { _id: 2, notes: 3 } ]
medirecord> db.clinical_notes.aggregate([ { $unwind: "$symptoms" }, { $group: { _id: { diagnosis_
code: "$diagnosis_code", symptom: "$symptoms" }, count: { $sum: 1 } } }, { $sort: { count: -1 } }
, { $limit: 8 } ])
[
  { _id: { diagnosis_code: 'J06', symptom: 'fever' }, count: 2 },
  { _id: { diagnosis_code: 'J45', symptom: 'wheezing' }, count: 2 },
  { _id: { diagnosis_code: 'J06', symptom: 'sore throat' }, count: 2 },
  {
    _id: { diagnosis_code: 'J45', symptom: 'chest tightness' },
    count: 2
  },
  { _id: { diagnosis_code: 'K29', symptom: 'abdominal pain' }, count: 1 },
  {
    _id: { diagnosis_code: 'G43', symptom: 'sensitivity to light' },
    count: 1
  },
  { _id: { diagnosis_code: 'J44', symptom: 'wheezing' }, count: 1 },
  { _id: { diagnosis_code: 'G43', symptom: 'headache' }, count: 1 }
]
medirecord>
```

40. Web app

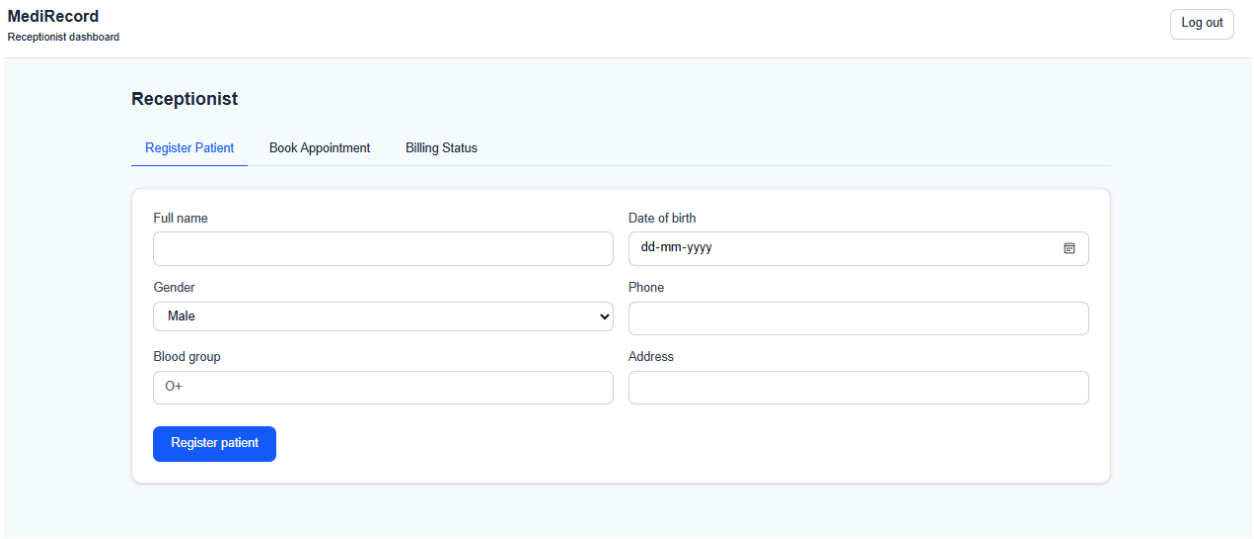


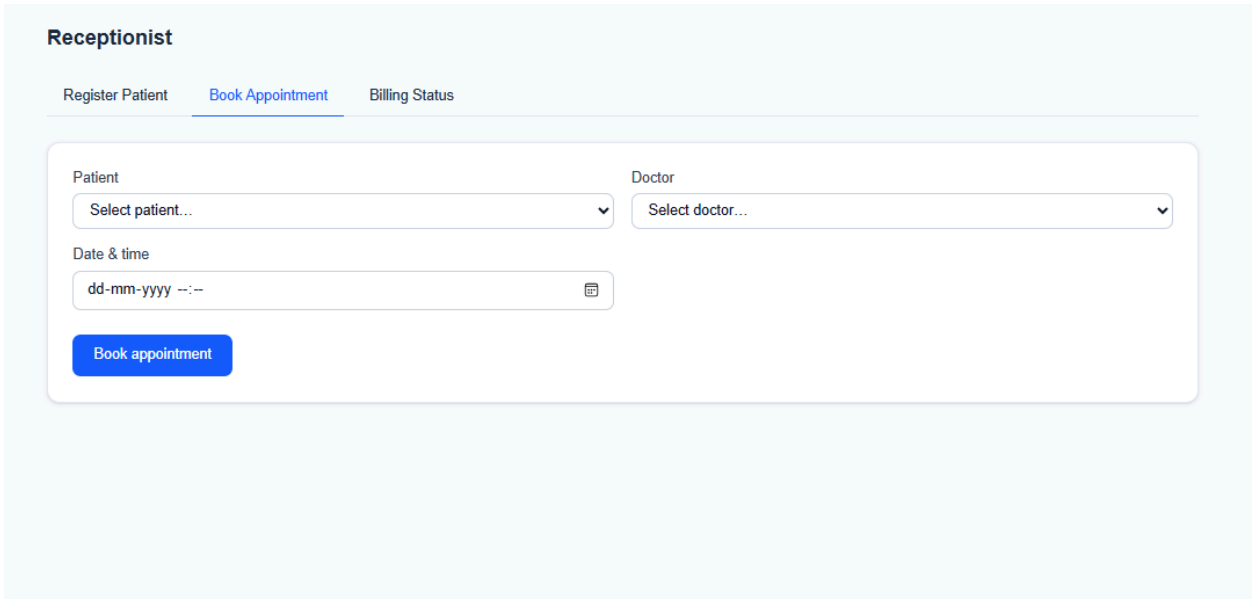
Admin:

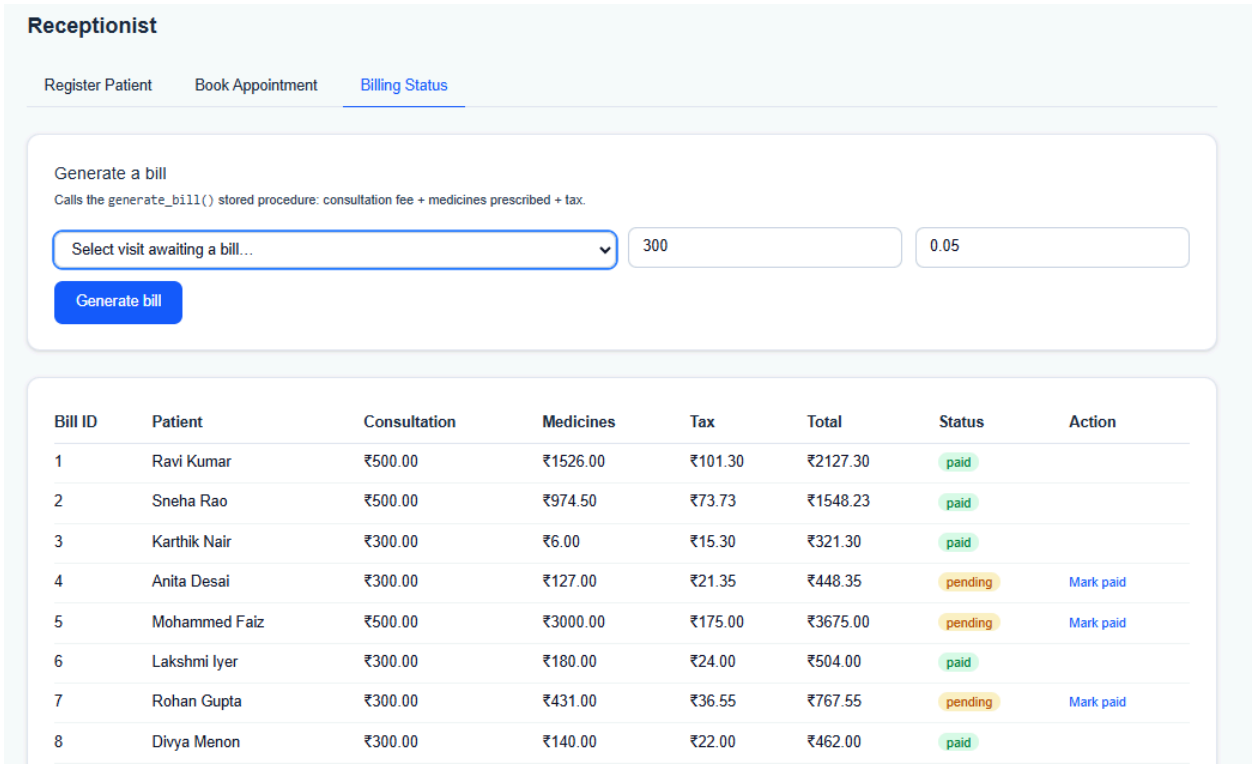




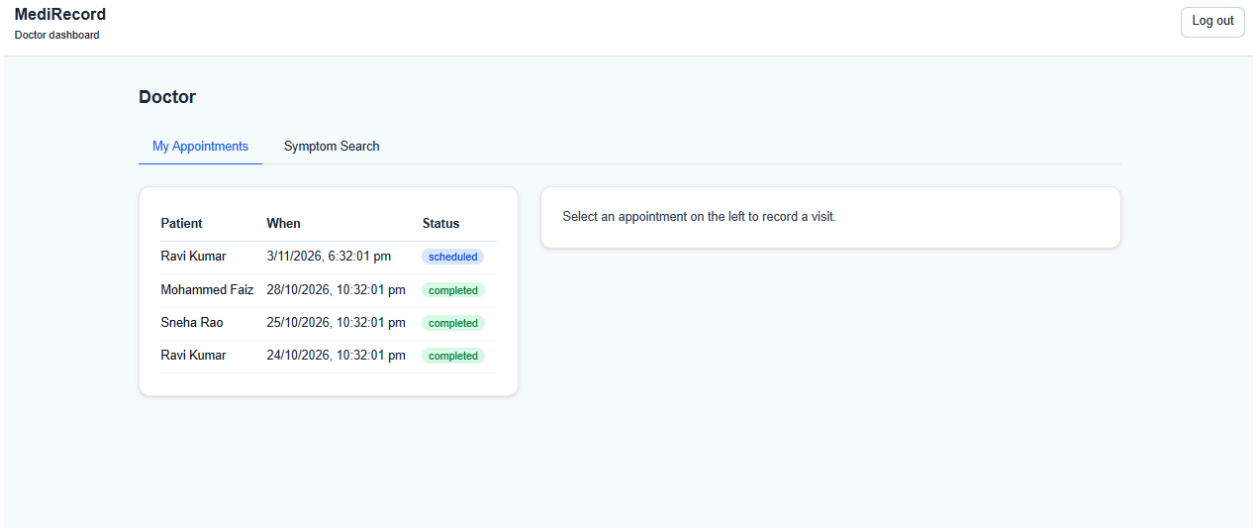
Receptionist:


MediRecord
Receptionist dashboard
Log out
Receptionist
Register Patient
Book Appointment
Billing Status
Full name
Date of birth
dd-mm-yyyy
Gender
Male
Phone
Blood group
O+
Address
Register patient


Receptionist
Register Patient
Book Appointment
Billing Status
Patient
Select patient…
Doctor
Select doctor…
Date & time
dd-mm-yyyy --:--
Book appointment


Receptionist
Register Patient
Book Appointment
Billing Status
Generate a bill
Calls the generate_bill() stored procedure: consultation fee + medicines prescribed + tax.
Select visit awaiting a bill…
300
0.05
Generate bill
Bill ID Patient Consultation Medicines Tax Total Status Action
1 Ravi Kumar ₹500.00 ₹1526.00 ₹101.30 ₹2127.30 paid
2 Sneha Rao ₹500.00 ₹974.50 ₹73.73 ₹1548.23 paid
3 Karthik Nair ₹300.00 ₹6.00 ₹15.30 ₹321.30 paid
4 Anita Desai ₹300.00 ₹127.00 ₹21.35 ₹448.35 pending Mark paid
5 Mohammed Faiz ₹500.00 ₹3000.00 ₹175.00 ₹3675.00 pending Mark paid
6 Lakshmi Iyer ₹300.00 ₹180.00 ₹24.00 ₹504.00 paid
7 Rohan Gupta ₹300.00 ₹431.00 ₹36.55 ₹767.55 pending Mark paid
8 Divya Menon ₹300.00 ₹140.00 ₹22.00 ₹462.00 paid

Doctor:


MediRecord
Doctor dashboard
Log out
Doctor
My Appointments
Symptom Search
Patient When Status
Ravi Kumar 3/11/2026, 6:32:01 pm scheduled
Mohammed Faiz 28/10/2026, 10:32:01 pm completed
Sneha Rao 25/10/2026, 10:32:01 pm completed
Ravi Kumar 24/10/2026, 10:32:01 pm completed
Select an appointment on the left to record a visit.

Doctor

My Appointments | Symptom Search

Patient	When	Status
Ravi Kumar	3/11/2026, 6:32:01 pm	completed
Mohammed Faiz	28/10/2026, 10:32:01 pm	completed
Sneha Rao	25/10/2026, 10:32:01 pm	completed
Ravi Kumar	24/10/2026, 10:32:01 pm	completed

Ravi Kumar — completed
3/11/2026, 6:32:01 pm

Blood pressure	Temperature	Pulse	Diagnosis code
120/80	98.0 °F	82 bpm	J45

Clinical notes (MongoDB)

signs of asthma , COPD

chest tightness | wheezing

Prescription

No medicines prescribed.

Bill

Not billed yet.

Doctor

My Appointments | Symptom Search

Patient	When	Status
Ravi Kumar	3/11/2026, 6:32:01 pm	scheduled
Mohammed Faiz	28/10/2026, 10:32:01 pm	completed
Sneha Rao	25/10/2026, 10:32:01 pm	completed
Ravi Kumar	24/10/2026, 10:32:01 pm	completed

Mohammed Faiz — completed
28/10/2026, 10:32:01 pm

Blood pressure	Temperature	Pulse	Diagnosis code
136/88	98.8 °F	92 bpm	J44

Clinical notes (MongoDB)

Long-term smoker with chronic productive cough and breathlessness on climbing stairs. Expiratory wheeze on auscultation. Diagnosis revised to COPD.

shortness of breath | chronic cough | wheezing

Prescription

Medicine	Dosage	Frequency	Days
Salbutamol Inhaler	2 puffs	3 times a day	30
Azithromycin 500mg	1 tablet	once a day	5

Bill

Total ₹3675.00 (consultation ₹500.00 + medicines ₹3000.00 + tax ₹175.00) pending

Doctor

My Appointments | Symptom Search

Search past clinical notes by symptom or keyword (MongoDB text search), joined back to each case's full record in MySQL.

chest

Search

Ravi Kumar — visit #1 — score 1.28

Diagnosis: J45 · Doctor: Arvind Rao

Patient reports wheezing at night and chest tightness on exertion. Known asthma since childhood. Reduced peak flow. Advised to avoid dust exposure.

chest tightness | wheezing | shortness of breath

Sneha Rao — visit #2 — score 0.75

Diagnosis: J45 · Doctor: Arvind Rao

Dry cough with wheezing after exercise, worse in cold weather. Mild bronchospasm on auscultation. Suspected exercise-induced asthma.

wheezing | dry cough | chest tightness